

Specificity and “Non-Derived Environment Blocking” in Logical Phonology*

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1. Introduction

Logical Phonology (LP, e.g., Bale et al. 2014; Bale and Reiss 2018; Leduc et al. 2024; Reiss 2021; Gorman and Reiss 2026) deconstructs the arrow ‘ $\rightarrow$ ’ of traditional phonological rules into several distinct operators. So far, treating segments as *sets* of (valued) features, most LP work has focused on intrasegmental changes effected by rules, and has proposed two simple operations, unification and subtraction, from which all such rules are constructed. Using these two operations, LP reverses the bias of many older models that required a diacritic on feature-filling rules, rules that cannot change feature values on segments, but only fill in absent values. This reversal is accomplished by treating feature-*filling* processes as fundamental, each modeled by a single unification-based rule. Formalizing an old idea, LP models feature-*changing* processes, not as primitive functions, but by a sequence of a subtraction rule and a unification rule. A result of this simple LP approach has been the reduction of many patterns that appear to be subject to lexically or morphologically conditioned “exceptions” down to purely phonological processes based on simple LP rules and a judicious use of underspecified lexical entries.

In this paper, I focus on LP accounts of so-called “non-derived environment blocking” (NDEB) of phonological rules. LP has no constraints on representations or rule application and no blocking mechanisms. In order to illustrate the basic mechanism of LP for achieving the effects of feature-filling rules, I translate into the LP framework an insightful Optimality Theoretic analysis offered by Inkelas (2000) to replace an earlier NDEB analysis for Finnish (Kiparsky 1993). This first discussion uses only unification rules, and adopts Inkelas’s idea that “exceptionality” among segments in a rule target position, the failure of some segments to undergo a rule’s effects, is the manifestation of inalterability as prespecification (IaP) vs. “normal” target mutability as underspecification.

Next, I offer a feature-changing alternative to the first analysis of Finnish, building on some conceptual tools introduced to deal with a puzzle in Barrow Inupiaq. This allows me to introduce subtraction rules in LP, and it also allows me present the LP extension of Inkelas’s IaP insight to segments in *trigger* position, segments that constitute rule environments (see Gorman and Reiss 2026 for extensive discussion).

Finally, I turn to another putative case of NDEB involving deletion of velar stops in Turkish. I show how standard feature logic, along with a third LP operator, introduced in

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Bale and Reiss (2018) and Reiss (Submitted), the one that effects full segment insertion and deletion, provides a simple, albeit counterintuitive, narrow phonological solution, that, once again, vitiates the NDEB proposals. This result is striking in light of Inkelas’s (2014:271) conclusion that a purely phonological analysis is not possible, that Turkish velar deletion “turns out to be highly morphologically and phonologically conditioned” and that a solution based on underspecification “would be far-fetched and un insightful” (Inkelas 2000:25).

2. Non-derived environment blocking (NDEB)

In Finnish (1) the root for ‘water’ has a final [e~i] alternation, but the root for ‘taxi’ has non-alternating [i]. It thus appears that Finnish raises /e/ to [i] word-finally.

- (1) a. *vetenä* ‘water-ESSIVE.SG vs. *vesi* ‘water-NOM.SG from /vete/
 b. *taksina* ‘taxi-ESSIVE.SG vs. *taksi* ‘taxi-NOM.SG from /taksi/

The forms in (1) also suggest that in *vesi*, the underlying /t/ surfaces as [s] before the [i] that results from the raising rule. Informally, there appears to be a rule ‘s → t / ___i’. Underlying /i/ vowels, not from /e/, also trigger this assibilation, as in the root ‘want’ (2).

- (2) *halut-a* ‘want-INFINITIVE’ vs. *halus-i* (/halut-i/) ‘want-PAST’

In *vesi*, the change of *t* to *s* occurs **within a morpheme**. It seems that underlying /t/ surfaces as [s] before the [i] that results from the raising rule—the raising process *feeds* the assibilation in the phonological derivation. In contrast, in *halusi* the change occurs **across a morpheme boundary**. The phonological rule clearly applies after the morphological derivation that combines the verbal root with the past tense suffix, since it is the concatenation of morphemes that creates the *ti* sequence in /halut-i/.

A challenge arises when we note that the sequence *ti* surfaces unchanged in *tila* and *äiti* (3) when originating from a morpheme internal sequence that arises neither from the concatenation of morphemes (as in *halusi*) nor from the effects of another rule (as in *vesi*).

- (3) Assibilation “blocked” in lexical *ti*:
 a. *tila* ‘room-NOM.SG’ b. *äiti* ‘mother-NOM.SG.’

Because the two environments where the change does occur do not obviously form a natural class—one refers to phonological derivation and the other to morphological derivation—Kiparsky (1993) proposes a general phonological rule that is blocked from applying in **non-derived** environments, whence NON-DERIVED ENVIRONMENT BLOCKING (NDEB). That is, the two environments where assibilation occurs are posited to actually constitute a single environment: a sequence arising from morpheme concatenation and a sequence arising from the application of a prior phonological rule can be grouped together as representing “derived environments,” and the process is blocked in non-derived environments.

For anyone committed to a modular view of grammar, such a solution equating two distinct notions of “derivation” will be unappealing. Thus, it is not surprising that many alternatives have been explored. Rasin (2015) surveys approaches to NDEB, including Lexical Phonology and OT proposals:

Most previous literature on NDEB has tackled the challenge by introducing special mechanisms to distinguish derived from nonderived environments. Some representative examples are the Strict Cycle Condition (Mascaró 1976), identity rules on lexical items combined with the Elsewhere Condition in Lexical Phonology and Morphology (Kiparsky 1982), a distinction between “new” and “old” structure in Comparative Markedness (McCarthy 2003), and morpheme coloring in Colored Containment (van Oostendorp, 2007).

Rasin’s own solution for Finnish is similar to previous ones that appeal to both underspecification and the contrast of feature-filling and feature-changing processes. However, his solution still requires enhancements to pure phonology, like Morpheme Structure Rules that apply to individual morphemes and constraints on the lexical segment inventory. Such mechanisms are anathema to the LP program, so I explore alternatives.

3. Inkelas’s feature-filling approach for Finnish

Inkelas’s (2000) proposal reduces to the following ideas. Finnish shows surface [t]’s that alternate with [s] and surface [t]’s that do not alternate. Given an impressive range of arguments that underspecification cannot be prohibited, she points out that an elegant way to encode the distinction between alternating and non-alternating *t* is to treat the non-alternating one as underlyingly specified, or *prespecified* as –CONTINUANT, whereas the alternating one is underspecified with respect to CONTINUANT. The former segment is denoted /t/, and the latter, which surfaces as either [t] or [s], depending on context, is denoted /T/. Following Inkelas and Cho (1993), the LP literature refers to this situation as Inalterability as Prespecification (IaP). In (4) I indicate the featural contrast between the prespecified, non-alternating /t/ of *tila* and the underspecified, alternating /T/ of *vetenä*, *vesi*.

	/t/	/T/
(4) CONTINUANT	–	
CORONAL	+	+
VOICE	–	–
<i>etc.</i>		

Inkelas proposed that assibilation is the result of purely feature-filling effects, so underlying /T/ can be affected, without affecting underlying /t/. A non-LP grammar, using diacritically specified feature-filling rules, or analogous Optimality Theoretic mechanisms, ensures that /t/ is immune to assibilation, but /T/ is not.

- (5) Feature-filling: Voiceless coronal stop becomes +CONTINUANT / ___ i
- /Ti/ $\rightsquigarrow$ [si] in /halut-i/ $\rightsquigarrow$ [halusi] (feature-filling)
 - /ti/ $\rightsquigarrow$ [ti] in /tila/ $\rightsquigarrow$ [tila] (feature-changing is blocked)

Underspecified /T/ is affected by the feature-filling rule, but /t/ is not.

4. Feature filling in LP

The challenge for formulating feature filling rules has been noted for decades (e.g., Lees 1961:12–14, Lightner 1971:236, Archangeli 1988, Reiss 2003). The generative literature for the most part assumes that feature changing rules are basic, and that special mechanisms are needed, either rule diacritics or special representational devices, to ensure that a rule applies in only a feature filling manner (e.g., Kiparsky 1985; Inkelas and Cho 1993; Buckley 1994; Inkelas 1995). The problem is that a feature-filling rule must make reference to an *absence* of information, for example, the absence of a value for CONTINUANT in the rule target class. LP has found a way to address this problem by making use of *unification*, a partial operation. I merely sketch the system here because this approach has been described in detail in a range of published work (e.g., Bale et al. 2014; Bale and Reiss 2018; Reiss 2021; Leduc et al. 2024; Gorman and Reiss 2026).

In LP we can handle segment internal changes effected by rules by treating segments (basically) as sets of (valued) features, like +ROUND, and so on. Feature filling is then implemented by rules based on the logical operation of unification, which combines the valued features of a segment (a set of features) from the target class with the valued features from a set representing the rule’s structural change.

LP aspires to provide a precise characterization of what is a possible rule. Part of this characterization is that rule targets must be expressed as natural classes of segments. For example, a rule targeting nasal consonants targets every segment which contains the features +NASAL and –SYLLABIC. Equivalently, such a rule targets every segment which is a superset of the set of features {+NASAL, –SYLLABIC}. Such natural classes can contain several segments which all share a conjunction of properties, as in the case of the English nasals, but they can also be restricted to a single segment by specifying all the features that characterize a segment in the natural class description. For example, in English, only /m/ is a superset of set of features {+NASAL, –SYLLABIC, +LABIAL, –CORONAL, –DORSAL, ...}. In order to make a distinction between the segment /m/, which is a set of valued features, and the natural class containing just the segment /m/, which is a set of sets of valued features, we distinguish brackets as in (6).

(6) The segment /m/ vs. the natural class containing only /m/

$$\begin{array}{ccc}
 \text{segment /m/} & & \text{natural class \{m\}} \\
 \left\{ \begin{array}{c} + \text{ NASAL} \\ - \text{ SYLLABIC} \\ + \text{ LABIAL} \\ - \text{ CORONAL} \\ - \text{ DORSAL} \\ \dots \end{array} \right\} & \neq & \left[\begin{array}{c} + \text{ NASAL} \\ - \text{ SYLLABIC} \\ + \text{ LABIAL} \\ - \text{ CORONAL} \\ - \text{ DORSAL} \\ \dots \end{array} \right]
 \end{array}$$

The problem of feature-filling rules arises when the characterization of one segment is a subset of the characterization of another. This is the case with Inkelas’s proposed /T/

and /t/, where the first is a subset of the second. Let the symbol $\textcircled{t}$ denote the list of features in the set corresponding to /t/. If we make a natural class using this set of features we get $[\textcircled{t}]$ which is just the singleton natural class {t}. However, the natural class denoted by $[\textcircled{T}]$ is a set containing at least /t/ and /T/, because each of these segments is a superset of /T/—recall that a basic theorem of set theory is that every set is a superset of itself. In fact, the smallest natural class containing /t/ and /T/ must also contain /s/, so the natural class is {t, T, s}. Thus, there is no way to define a natural class containing /T/ but not /t/. Of course this means that a rule cannot target /T/ without targeting /t/ and /s/, unless we just stipulate that that is the case. But we don't want a model that allows rules to contain such stipulation. LP offers a formal model of rules in which targets are *always* natural classes, not the informal notion (e.g., Hayes 2009) that natural classes occur “in most instances” (p. 43) or as a “pervasive tendency” (p. 71).

5. Exceptionality as *target prespecification*: ‘NDEB’ as pure phonology

The LP implementation of Inkelas's idea is to use a rule based on a unification operator, $\sqcup$. Unification in LP works like this: The unification of two sets, $A \sqcup B$ is equal to their union $A \cup B$ if that union is consistent, if it does not contain opposite values for any feature; otherwise, the rule application is vacuous and the output of the rule is A .

(7) Feature filling assibilation by unification

$$\begin{bmatrix} -\text{VOICED} \\ +\text{CORONAL} \\ -\text{SONORANT} \end{bmatrix} \sqcup \{+\text{CONTINUANT}\} / __ \begin{bmatrix} -\text{BACK} \\ +\text{HIGH} \\ -\text{ROUND} \end{bmatrix}$$

Note that both the target and the environment of the rule are characterized via natural classes in square brackets, since they refer to sets of segments. In contrast, the structural change refers just to a set of features. Also, note that the target class necessarily includes /t, T, s/. Using the circled notation, we can write the same rule a bit less formally (8).

(8) Feature filling assibilation by unification

$$[\textcircled{T}] \sqcup \{+\text{CONTINUANT}\} / __ i$$

Let's now consider the three members of the target class in a position before /i/:

(9) Yield of assibilation rule

- a. /t/ $\sqcup \{+\text{CONTINUANT}\}$ is vacuous, since /t/ contains the conflicting feature $-\text{CONTINUANT}$. The output is /ti/. This is why /tila/ $\rightsquigarrow$ [tila].
- b. /T/ $\sqcup \{+\text{CONTINUANT}\}$ inserts a feature into /T/, and yields [si], seen in the mapping /haluT-i/ $\rightsquigarrow$ [halusi]. This is feature filling.
- c. /s/ $\sqcup \{+\text{CONTINUANT}\}$ is vacuous unification, because the union of sets is vacuous. Nothing changes when unifying /s/ with a feature it already contains: /si/ $\rightsquigarrow$ [si] as in /taksi/ $\rightsquigarrow$ [taksi].

In other words, Inkelas proposes that some exceptionality—failure to undergo a rule—is due to (full, lexical) *prespecification*. Only underspecified targets are affected by feature-filling. The advantage of Inkelas’s approach is that it provides a purely phonological analysis using independently motivated notions, such as underspecification, and it makes unnecessary the use of lexical exceptions or bogus natural classes. Following Inkelas, we refer to the prespecified segments that fail to be affected by feature-filling processes “inalterable,” whereas the affected segments are called “mutable.” We have found that Inkelas’s Inalterability as Prespecification IaP insight, implemented in LP, allows us to trivially reanalyze case after case of supposed lexical or morphological conditioning of segmental alternations as narrow phonological processes expressed with the austere LP toolkit (e.g., Bale et al. 2014; Gorman and Reiss 2026).

6. Set subtraction rules

In addition to the unification operator already introduced, LP uses one more set-theoretic rule type. With the simple assumption that segments are sets of features, neutralization of, say, /t/ and /d/ to the coronal stop /D/, unspecified for VOICED, can be modeled via the normal set subtraction operator: $/t, d/ \setminus \{+VOICED, -VOICED\}$ yields /D/ in both cases. Just like unification, subtraction can apply vacuously. For example, $/d/ \setminus \{-VOICED\}$ vacuously yields /d/, because there is no $-VOICED$ specification to remove from a /d/. Note, also, that subtraction can change a segment from inalterable to mutable in the course of a derivation, as in the case of mapping /d/ to /D/. Thus, feature-*changing* processes, for example, changing a voiced /d/ to a voiceless /t/, do not require a separate operator. Feature-changing can be modeled as a sequence of a subtraction rule, followed by a unification rule, an idea with a long history, as mentioned above (e.g., Harris 1984; Mester and Itô 1989; Poser 1993; Wiese 2000; McCarthy 2007; Samuels 2011).¹ I make use of this idea below.

7. Exceptionality as *trigger underspecification*

The IaP distinction between two types of potential targets requires no enhancements to phonology beyond admitting underspecification. In addition to the many arguments Inkelas has provided for the necessity of underspecification, it should also be noted that allowing underspecification involves a simplification to phonology, namely doing without a stipulation that segments must be specified for each feature (Reiss 2012). Therefore, it is important to note that this same idea can also be extended to differentiate segments that

¹Some earlier work implicitly or explicitly noted the elegance of breaking feature-changing down like this, but in autosegmental models the feature-filling aspect is typically identified with spreading. In LP, which is a component of Substance Free Logical Phonology, there is no distinction between spreading and “insertion” of a feature. In other words, we reject the special status of assimilation. This is one of the fundamental principles of substance free phonology as originally formulated by the “Concordia School” (e.g., Hale and Reiss 2000, 2008; Reiss 2018; Volenec and Reiss 2020; Reiss and Volenec 2022). The idea is encoded by Dabbous et al. (2021) as the Substance-Freeness of Structural Change: Features changed by rule application need not be present in the rule’s structural environment.

are potential rule *triggers*. This possibility, not exploited by Inkelas and colleagues, can be illustrated here with a simple case, with more provided by Gorman and Reiss (2026).

Barrow Inupiaq has three surface vowels: [i, a, u]. As discussed by Buckley (1994), Archangeli and Pulleyblank (1994:§2.2.2), and Dresher (2009:§7.2.1), among others, it appears to be the case that there is a rule palatalizing coronal sonorants after /i/, as in (10a), where the stem final /i/ in ‘wound’ causes the palatalization of segments that show up unpalatalized elsewhere. Interestingly, as in (10b), there are even cases of stems ending in /i/, such as ‘place’ where palatalization fails to appear. The LP proposal, basically adopted from the earlier discussions, is that the surface vowel [i] corresponds to two distinct underlying vowels. The so-called “strong” *i*’s are, in LP terms, “catalytic” and trigger palatalization of a following coronal sonorant (10a) but “weak” *i*’s are inert, or “quiescent,” and do not (10b).

- (10) Palatalization in Barrow Inupiaq (Kaplan 1981:§3.22)
- a. iki ‘wound’ iki-**ɬ**u ‘and a wound’ iki-**ɲ**ik ‘wounds’
 - b. ini ‘place’ ini-**ɬ**u ‘and a place’ ini-**n**ik ‘places’

I propose that weak *i*, denoted /I/, as in the final vowel of *ini*, is underspecified relative to strong *i*, the final vowel of *iki* (11).

- (11) Underlying vowels of Barrow Inupiaq
- | | strong <i>i</i> = i | weak <i>i</i> = I | a | u |
|------|---------------------|-------------------|---|---|
| HIGH | + | + | – | + |
| BACK | – | | + | + |

I then derive palatalization via a two-step feature-changing process involving a subtraction rule (12) and a unification rule (13).

- (12) First rule removes +ANTERIOR from coronals sonorants after /i/
- $$\left[\begin{array}{c} +\text{CORONAL} \\ +\text{SONORANT} \end{array} \right] \setminus \{ +\text{ANTERIOR} \} / \left[\begin{array}{c} +\text{HIGH} \\ -\text{BACK} \end{array} \right] \text{ —}$$

Note that strong *i*, but not weak *i* (I), is a trigger for the rule—/I/ is not a member of the natural class $[\textcircled{\text{i}}]$.

- (13) Next rule adds –ANTERIOR to coronals
- $$[+\text{CORONAL}] \sqcup \{ -\text{ANTERIOR} \}$$

Rule (13) needs no context, since it will only fill in a –ANTERIOR value on coronals that lack a value.² Finally, we need to ensure the absolute neutralization at the surface of weak

²The target also need not be specified as +SONORANT, since only sonorants will have lost a +ANTERIOR value by the previous rule. I am using these compact rules, not because I believe that rules are necessarily represented in the most economical form possible (see Hale and Reiss 2008; Gorman and Reiss In prep for discussion), but rather in order to illustrate the workings of unification failure in LP.

and strong *i*'s, which is accomplished with a unification rule. The rule target is given using a featural representation in (14a), and with the circled notation in (14b). In both cases, the target natural class is $\{i, I, u\}$, but only $/I/$ can be changed by the rule, since unification is vacuous for $/i/$, and it fails for $/u/$ —yielding two types of vacuous rule application.

$$(14) \quad \text{a. } [+HIGH] \sqcup \{-BACK\} \qquad \text{b. } [\textcircled{I}] \sqcup \{-BACK\}$$

At the point in the derivation when palatalization occurred, weak *i* = $/I/$ was quiescent—it could not trigger the palatalization process. Actually, it is worthwhile to point out that there is no palatalization rule *per se* in this analysis. The segment $/I/$ is quiescent with respect to the subtraction rule (12), and the unification rule (13) has no triggering segments, so the status of $/i/$ vs $/I/$ is not relevant at this point in the derivation. This is consistent with the LP views that traditional terms like palatalization and assimilation are “taxonomic artifacts” in the sense of Chomsky (2000:8), and with the notion of substance-free phonology (Hale and Reiss 2008; Reiss 2018).

8. Trigger underspecification can also replace NDEB in Finnish

For purposes of our analysis of Finnish, we assume that the language has the same distinction as Barrow Inupiaq, with two *i*'s, strong, fully specified, catalytic $/i/$, and weak, partially specified, quiescent $/I/$. Since $/i/$ is a superset of $/I/$, the former can constitute a natural class without the other: $\{i\}$ is a definable natural class. Therefore, rules of Finnish can be triggered by the natural class defined by all the features of $/i/$, denoted $[\textcircled{i}]$. Given this distinction between catalytic and quiescent vowels in the target position of the assibilation rule, there is no need to distinguish two segments $/t/$ vs. $/T/$ in target position, as we did in the previous analysis, following Inkelas's IaP approach. Either of these representations, $/t/$ or $/T/$, can be used to generate the correct output. Since our goal in this paper is to illustrate the mechanisms of LP, I present a feature-changing analysis with underlying $/t/$. Features of relevant vowels appear in (15).

(15) Finnish vowel features

	$/e/$	$/E/$	$/i/$	$/I/$
BACK	—	—	—	
HIGH	—		+	+

We need five “primitive” LP rules for Finnish raising and assibilation. Recall that some underlying *i*'s as well as some *i*'s derived from $/e/$, as in *vesi*, trigger assibilation. We thus need to change the word final $[e]$ of $/vete/$ to $[i]$. The first step is to remove that vowel's specification as $-HIGH$:

$$(16) \quad [\textcircled{e}] \searrow \{-HIGH\} / __ \% \qquad (e \rightarrow E \text{ at the end of word})$$

The affected vowels thus become mutable by the next rule.

$$(17) \quad [\textcircled{E}] \sqcup \{+HIGH\} \qquad (E \rightarrow i)$$

We now have some derived /i/'s which may occur after an underlying /t/, as in /vete/ $\rightsquigarrow$ /veti/. Of course, the target class of the rule, as written, includes /i, e, E/ (but not /I/); however, application is vacuous to all except for the derived vowel /E/.

The next step is to make those /t/'s that occur before an (underlying or derived) /i/ mutable via subtraction of $-\text{CONTINUANT}$.

$$(18) \quad [\textcircled{t}] \setminus \{-\text{CONTINUANT}\} / ___ \left[\begin{array}{c} -\text{BACK} \\ +\text{HIGH} \\ -\text{ROUND} \end{array} \right] \quad (t \rightarrow T \text{ before } i)$$

In the target, I used $\textcircled{t}$ because there are no /T/'s until this point in the derivation, but $\textcircled{T}$ would also yield the correct result. In the environment, I could have used $[\textcircled{i}]$, since /i/ is the only vowel that is a superset of $\{-\text{BACK}, +\text{HIGH}, -\text{ROUND}\}$.

However, the next, feature-filling rule must target $[\textcircled{T}]$, since $[\textcircled{t}]$ would target only /t/, and there are no /t/'s in the relevant environment.

$$(19) \quad [\textcircled{T}] \sqcup \{+\text{CONTINUANT}\} \quad (T \rightarrow s)$$

Of course, the target class of this rule includes /t, T, s/. After application of rule (19), the derived /T/'s have all become $+\text{CONTINUANT}$ [s].³

Finally, remaining tokens of the quiescent /I/ vowel, like the Barrow “weak” *i*, are neutralized with the strong *i*.

$$(20) \quad [\textcircled{I}] \sqcup \{-\text{BACK}\} \quad (I \rightarrow i)$$

Note that we are ignoring the other Finnish high vowels, /y, u/, for sake of the exposition.

This second Finnish analysis is again purely phonological, like that offered by Inkelas, and so there is still no need to posit NDEB. Unlike Rasin’s proposal, there is no need for Morpheme Structure Constraints or constraints on segment inventories. Everything can be handled with unification and subtraction rules.

9. Full segment insertion and deletion in LP

LP is a work in progress. All intrasegmental processes appear to be amenable to analysis via subtraction and unification rules, but there are other kinds of phenomena that are still being studied, such as rules that refer to syllable structure, tonal patterns, or rules that effect metathesis. One domain that has been explored somewhat is that of full segment insertion and deletion. In order to account for such rules we first enhance our notion of segments to include association of feature sets to X-slots, timing units.

³We ignore other redundant changes like the change from $-\text{STRIDENT}$ to $+\text{STRIDENT}$.

- (21) X-slot representation of /i/

$$\begin{array}{c} \text{X} \\ | \\ \left\{ \begin{array}{l} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ +\text{HIGH} \\ -\text{LOW} \\ +\text{ATR} \\ \dots \end{array} \right\} \end{array}$$

Next we introduce the null segment symbol ε , which replaces the overburdened null or empty set symbol $\emptyset$ traditionally used in the literature. This is a crucial move, because deletion of a whole segment should not be confused with removal of all of a segment's features. Removal of all members of a set yields an empty set, which is not the same thing as nothing—just as an empty box is a box and not nothing. A segment consisting of an empty set associated to an X-slot is a fully underspecified segment, sometimes identified with a maximally “unmarked” segment like the vowel /ə/.

- (22) An empty segment set is not the same as the absence of a segment

$$\begin{array}{c} \text{X} \\ | \\ \{ \} \end{array} \neq \varepsilon$$

Combining this notation with the square brackets for natural classes, (23a) is a rule that deletes any member of the class of nasals at the end of a word, whereas rule (23b) inserts a segment that is specified only as +NASAL at the end of a word.⁴

- (23) Word-final nasal deletion vs. insertion of underspecified nasal.

$$\begin{array}{cc} \text{a.} & \begin{array}{c} \text{X} \\ | \\ [+NASAL] \end{array} & \mapsto & \varepsilon & / __\% \\ \text{b.} & & & & \varepsilon & \mapsto & \begin{array}{c} \text{X} \\ | \\ \{+NASAL\} \end{array} & / __\% \end{array}$$

It is obviously important to pay attention to the brackets!

10. Pessimism about “NDEB” involving segment deletion

We now have all the pieces in place to address another putative case of NDEB. Inkelas (2000, 2014) discusses the deletion of velars /k, g/ in Turkish between vowels. The nominative form [sokak] ‘street’ shows that [k] can surface between vowels. However, the stem final /k/ appears to be deleted when the dative ending is added, as in [sokaa]. The other forms in (24) show further examples with /k/ and also /g/. For example, the stem final /g/

⁴It should be clear that a rule deleting only the segment that rule (23b) inserts cannot be formulated (see Reiss Submitted).

of [katalog] does not show up in the dative, but there is an intervocalic [g] in [vagon]. Because the disappearing velars occur at the end of the stem when another morpheme follows, Inkelas discusses Turkish velar deletion as another potential case of NDEB—the rule only applies after the concatenation of the root and suffix morphemes.

(24) Turkish velar behavior

UR?	NOMINATIVE	DATIVE	
/sokak/	[so k ak]	[so k a-a]	‘street’
/bebek/	[bebe k]	[bebe-e]	‘baby’
/guguk/	[gu g uk]	[gu g u-a]	‘cuckoo’s call’
/katalog/	[katalo g]	[katalo-a]	‘catalog’
/mekanik/	[me k anik]	[me k ani-e]	‘mechanics’
/vagon/	[va g on]	[va g on-a]	‘wagon’

The leftmost column gives tentative UR’s.

Inkelas (2000) is pessimistic about an underspecification analysis for Turkish velars:

Velar Deletion deletes whole segments. One might attempt various strategies for featural prespecification of nonalternating /k, g/ and underspecification of alternating /k, g/, but since the alternation itself is not featural, these attempts would be far-fetched and un insightful.

Inkelas revisited Turkish velar deletion on several occasions, and while she ends up rejecting NDEB, she remains pessimistic about finding a phonological solution:

Although Turkish velar deletion appears to be a classic morphologically derived environment effect, closer inspection shows that no matter how “derived environment” conditioning is defined, Turkish velar deletion does not actually meet its description. The alternation turns out to be highly morphologically and phonologically conditioned. The fact that environments in which the rule does apply are derived is a side effect of its other morphological and phonological conditioning, not the explanatory factor. [Inkelas 2014:271]

I now present a simple LP solution that forces us to confront a prejudice inherent in Inkelas’s rejection of an underspecification analysis.

11. Exceptionality as *target underspecification*! The Logical Phonology solution

The LP solution for Turkish velars is to just trust natural class reasoning. I propose that Turkish has two velars that surface as [k] and two that surface as [g]. For each pair, let’s assume that one member is fully specified, so there are /k, g/, and one member is underspecified for some feature, say, DORSAL.⁵ Thus, Turkish has /k, K/ and /g, G/, where the second member of each pair is underspecified for DORSAL.

⁵Of course, it is important to consider how to determine *which* feature is the underspecified one, but we leave that aside for now.

		/k/	/K/	/g/	/G/
(25)	DORSAL	+		+	
	VOICED	–	–	+	+

Obviously, /k/ is a superset of /K/ and /g/ is a superset of /G/. Natural class reasoning tells us that the class defined by the features of the relatively less specified segments must contain the more specified segments.

(26) Natural class relations for Turkish velars

- $[\textcircled{K}] = \{k, K\}$ and $[\textcircled{G}] = \{g, G\}$
- $[\textcircled{k}] = \{k\}$ and $[\textcircled{g}] = \{g\}$

We can formulate a natural class that contains /k, g/ to the exclusion of /K, G/ by specifying +DORSAL and leaving VOICED unspecified. For convenience, let's use Γ for the relevant set of features, although there is no such segment in Turkish—we are using Γ only for a partial description, to characterize a natural class.

Then, the target of the deletion rule would be $[\textcircled{\Gamma}] = \{k, g\}$. Since we are focussing here on the velar stops versus all other segments, we can, for convenience reduce the natural class characterization to a single feature and write the target class as [+DORSAL], but I prefer $[\textcircled{\Gamma}]$ since it is consistent with LP views concerning the richness of rule specification (Gorman and Reiss In prep). The segment deletion rule can then be formulated purely phonologically as in (27).

(27) Velar Deletion Rule

$$[\textcircled{\Gamma}] \mapsto \varepsilon / [+SYLLABIC] \text{ ___ } [+SYLLABIC]$$

Counterintuitively, the alternating segments, the ones that sometimes delete, must be the fully specified ones, because fully specified segments can be targeted without targeting relatively underspecified ones, but not vice versa. This result just follows from standard natural class reasoning and cannot be dismissed just because it seems weird: “[I]f you believe P , and you believe that P entails Q , then even if Q seems more than a little odd, you have some intellectual obligation to take seriously the possibility that Q may be true, nonetheless” (Pylyshyn 1984:xxii). We now see how to revise the URs for the roots in (24). We must posit fully specified /k, g/ for the segments that alternate between [k] or [g] and the null segment ε . And we must posit underspecified /K, G/ for the *non-alternating* root-internal segments between vowels: e.g., /soKak, bebek, katalog, guGuk/.⁶

A potential objection to this analysis is that I have opportunistically chosen the URs so that the analysis “works”—all surfacing velars that are between vowels just happen to appear morpheme-internally and these are encoded as /K, G/, whereas the alternating ones

⁶I assume that the root-initial velars in /katalog/ and /guGuk/ are fully specified by virtue of my assumptions concerning the acquisition of phonological rules and representations that are discussed elsewhere (Hale and Reiss 2003; Gorman and Reiss In prep).

all occur at the end of roots. *Aren't we failing to account for a generalization? Aren't we arbitrarily replacing the NDEB analysis with another arbitrary one?* But these arguments are spurious. Every analysis wants to get the data right, so we always want to choose URs that work. Unlike recent trends (e.g., Tesar 2014) LP is “input driven,” with an overarching concern for underlying forms grounded on the assumption that the “relation between a phonemic system and the phonetic record . . . is remote and complex” (Chomsky 1964:p. 38). Further, I assume, consistent with what appears to be implicit in the literature, that children are *epistemically bounded* (in the sense of Fodor 1980:33f.) to pursue narrow phonological analyses before entertaining lexical, morphemic or morphosyntactic conditioning. LP gives us a narrow phonological account of Turkish velar deletion.

Finally, we have another argument. There are exceptions to the generalization that velars delete intervocalically in derived environments. The noun [orak] ‘sickle’ is [oraka] in the dative, and the noun [patalog] ‘pathologist’ is [pataloga] in the dative: these stem final velars do not delete intervocalically. Instead of positing NDEB as well as the possibility of lexical exceptions to NDEB, we accept the existence of underspecification and posit an underlying phonological distinction. These non-alternating, stem-final velars must be underspecified, that is /oraK, pataloG/, and thus they cannot be affected by a rule targeting the fully specified members of the class $[\Gamma]$, namely /k, g/. This is just traditional natural class logic.

Since, /K, G/ surface as /k, g/ when they are not deleted, a final fill-in rule is necessary. It is not possible to target only the underspecified segments, so a unification rule adding +DORSAL must target the whole class $\{k, K, g, G\}$. Let's use $[\mathcal{G}]$ to represent this class, as in rule (28).

(28) Default rule counterfeeds deletion: $[\mathcal{G}] \sqcup \{+DORSAL\}$

This rule turns /K, G/ into [k, g], respectively, but has no effect on any other segments. Representative derivations are provided in (29).

(29) Sample derivations

	‘street’	DAT	‘sickle’	DAT
UR	soKak	soKak-a	oraK	oraK-a
Rule (27):		soKaa		
Rule (28):	sokak	sokaa	orak	oraka
SR	sokak	sokaa	orak	oraka

This phonological solution requires one segment deletion rule and one unification rule.

Let's summarize the logic that brought us here. When a unification rule “casts a wide net” over a natural class like $\{t, D, d\}$, it is possible for the two kinds of vacuous application to ensure that just the underspecified segment /D/ is affected. In contrast, a segment deletion rule cannot apply vacuously to any representation that meets its structural description. If a deletion rule's net is cast wide, every member of the target class will be deleted. The target must be defined narrowly if only one segment is deleted.

12. Conclusion

I demonstrated how some putative NDEB cases can be reduced to narrow phonology in LP. First I translated Inkelas’s IaP insight and demonstrated how LP solves the problem of modeling feature-filling rules, building on the use of unification failure. I applied these tools to translate Inkelas’s rejection of NDEB in Finnish assibilation into a LP account, requiring no constraints on underlying forms and no reference by the grammar to the phoneme inventory, in contrast to work like Rasin (2015). I next extended the IaP insight to segments in rule environments, adding the notions of catalysis and quiescence of triggers to the inalterability and mutability that Inkelas introduced for targets. I applied these ideas to an alternative analysis of Finnish, but left open the question of which of the two LP analyses is preferable. Finally, I introduced the LP mechanism for full segment deletion and showed how it can be applied to another case of putative NDEB, Turkish velar deletion, which Inkelas suggested was *not* amenable to an underspecification analysis. Simply following standard natural class reasoning allowed for an analysis that is simple and robust in its ability to capture the basic “NDEB” patterns as well as apparent lexical exceptions to NDEB. This solution suggests that we must allow logic to trump intuitions and biases concerning which segment, a fully specified one or an underspecified one, is “more likely” to alternate with zero. The solution is not “far-fetched and un insightful,” but is rather a consequence of adopting an explicit theory of rules and representations and adhering to its logic.

References

- Archangeli, Diana. 1988. *Underspecification in Yawelmani phonology and morphology*. New York: Garland. Revised version of 1984 MIT dissertation.
- Archangeli, Diana, and Douglas Pulleyblank. 1994. *Grounded phonology*. MIT Press.
- Bale, Alan, Maxime Papillon, and Charles Reiss. 2014. Targeting underspecified segments: A formal analysis of feature changing and feature filling rules. *Lingua* 148: 240–253.
- Bale, Alan, and Charles Reiss. 2018. *Phonology: A formal introduction*. Cambridge, MA: MIT Press.
- Buckley, Eugene. 1994. Prespecification of default features: the two /i/’s of Kashaya. In *Nels 24: Proceedings of twenty-fourth annual meeting of the north east linguistic society*, 17–30.
- Chomsky, Noam. 1964. *Current issues in linguistic theory*. The Hague: Mouton.
- Chomsky, Noam. 2000. *New horizons in the study of language and mind*. Cambridge, MA, US: Cambridge University Press.
- Dabbous, Rim, Marjorie Leduc, Fatemeh Mousavi, Charles Reiss, and David Ta-Chun Shen. 2021. Satisfying long-distance relationships (without tiers): A strictly anti-local approach to phonology. Ms. URL: <https://ling.auf.net/lingbuzz/006329>.
- Dresher, B. Elan. 2009. *The contrastive hierarchy in phonology*. Cambridge University Press.
- Fodor, Jerry. 1980. Reply to Putnam. In *Language and learning: The debate between Jean Piaget and Noam Chomsky*, ed. Massimo Piattelli-Palmarini, 325–334. Harvard University Press.
- Gorman, Kyle, and Charles Reiss. 2026. Metaphony in Substance Free Logical Phonology. *Phonology*.
- Gorman, Kyle, and Charles Reiss. In prep. Maximal feature specification is feasible; minimal feature specification is not. *Glossa*. (Handout version from talk at GLOW 46 in Vienna at lingbuzz/007296).
- Hale, Mark, and Charles Reiss. 2000. “Substance abuse” and “dysfunctionalism”: Current trends in phonology. *Linguistic Inquiry* 31(1): 157–169.
- Hale, Mark, and Charles Reiss. 2003. The subset principle in phonology: Why the *tabula* can’t be *rasa*. *Journal of Linguistics* 39: 219–244.

- Hale, Mark, and Charles Reiss. 2008. *The phonological enterprise*. Oxford: Oxford University Press.
- Harris, James W. 1984. Autosegmental phonology, lexical phonology and Spanish nasals. In *Language sound structure*, ed. M. Aronoff and R. Oehrle, 67–82. Cambridge, MA: MIT Press.
- Hayes, Bruce. 2009. *Introductory phonology*. Malden, MA: Wiley-Blackwell.
- Inkelas, Sharon. 1995. The consequences of optimization for underspecification. In *Proceedings of the North East Linguistic Society 25*, ed. Jill Beckman, 287–302. University of Massachusetts, Amherst: GLSA.
- Inkelas, Sharon. 2000. Phonotactic blocking through structural immunity. In *Lexicon in focus*, ed. Barbara Stiebels and Dieter Wunderlich, 7–40. Akademie Verlag Berlin.
- Inkelas, Sharon. 2014. *The interplay of morphology and phonology*. Oxford: Oxford University Press.
- Inkelas, Sharon, and Young-Mee Yu Cho. 1993. Inalterability as prespecification. *Language* 69(3): 529–574.
- Kaplan, Lawrence D. 1981. *Phonological issues in North Alaskan Inupiaq*. Alaska Native Language Center.
- Kiparsky, Paul. 1982. Lexical morphology and phonology. In *Linguistics in the morning calm*, ed. Im-Seok Yang, 3–91. Hanshin.
- Kiparsky, Paul. 1985. Some consequences of Lexical Phonology. *Phonology Yearbook* 2: 85–138.
- Kiparsky, Paul. 1993. Blocking in nonderived environments. In *Studies in lexical phonology*, ed. Sharon Hargus and Ellen M. Kaisse, *Phonetics and Phonology*, vol. 4, 277–313. Boston: Academic Press.
- Leduc, Marjorie, Charles Reiss, and Veno Volenec. 2024. Votic vowel harmony in Substance Free Logical Phonology. In *Oxford Handbook of vowel harmony*. Oxford: Oxford University Press.
- Lees, Robert B. 1961. *The phonology of Modern Standard Turkish*. Indiana University Publications.
- Lightner, Theodore M. 1971. On Swadesh and Voegelin's 'A Problem in Phonological Alternation'. *International Journal of American Linguistics* 37(4): 227–237.
- Mascaró, Joan. 1976. Catalan phonology and the phonological cycle. Doctoral dissertation, MIT.
- McCarthy, John J. 2003. Comparative markedness. *Theoretical linguistics* 29(1-2): 1–51.
- McCarthy, John J. 2007. Slouching toward optimality: Coda reduction in OT-CC. *Phonological Studies (Journal of the Phonological Society of Japan)* 7: 89–104.
- Mester, R Armin, and Junko Itô. 1989. Feature predictability and underspecification: Palatal prosody in Japanese mimetics. *Language* 258–293.
- Poser, William J. 1993. Are strict cycle effects derivable? In *Studies in lexical phonology*, ed. Sharon Hargus and Ellen M Kaisse, vol. 4. San Diego: Academic Press.
- Pylyshyn, Zenon. 1984. *Computation and cognition: Toward a foundation for cognitive science*. Cambridge, MA: MIT Press.
- Rasin, Ezer. 2015. A rule-ordering theory of blocking in nonderived environments. In *Proceedings of the annual meetings on phonology*.
- Reiss, Charles. 2003. Deriving the feature-filling/feature-changing contrast: An application to Hungarian vowel harmony. *Linguistic Inquiry* 34(2): 199–224.
- Reiss, Charles. 2012. Towards a bottom-up approach to phonological typology. In *Towards a biolinguistic understanding of grammar*, ed. A.M. di Sciullo, 169–191. Philadelphia: John Benjamins.
- Reiss, Charles. 2018. Substance free phonology. In *The Routledge Handbook of Phonological Theory*, ed. Stephen J. Hannahs and Anna R. K. Bosch, 425–452. Oxford: Routledge.
- Reiss, Charles. 2021. Towards a complete Logical Phonology model of intrasegmental changes. *Glossa* 6(1): 107.
- Reiss, Charles. Submitted. Delete the rich: On the non-existence of 'weak' schwa deletion.
- Reiss, Charles, and Veno Volenec. 2022. Conquer primal fear: Phonological features are innate and substance free. *Canadian Journal of Linguistics* 67(4): 581–610.
- Samuels, Bridget D. 2011. *Phonological architecture: a biolinguistic perspective*. Oxford: Oxford University Press.
- Tesar, Bruce. 2014. *Output-driven phonology: Theory and learning*. Cambridge, U.K.: Cambridge University Press.
- Volenec, Veno, and Charles Reiss. 2020. Formal generative phonology. *Radical: A Journal of Phonology* 2: 1–148.
- Wiese, Richard. 2000. *The phonology of German*. Oxford: Oxford University Press.